

Estelle Chigot

PHD CANDIDATE IN COMPUTER VISION

Toulouse, France

🏠 echigot.github.io | 📧 echigot | 🌐 estelle-chigot | 🎓 Estelle Chigot

Experience

PhD Candidate

Toulouse, France

AIRBUS & ISAE-SUPAERO

2023 – June 2026

- Researcher within the vision-based navigation team at Airbus.
- Designed a diffusion-based style transfer method for synthetic-to-real domain adaptation using segmentation maps, with a focus on adverse weather conditions (CVIU 2025).
- Proposed an evolutionary algorithm-based optimization framework for style transfer pipelines (EvoStar 2026).
- Evaluated the impact of simulated data and domain adaptation methods for Airbus runway detection model (CAIP 2025).
- Applied and benchmarked synthetic-to-real style transfer models for runway images under adverse conditions.
- Taught computer vision (generative models), machine learning, object-oriented programming, evolutionary optimization and Bash/Python.
- Co-supervised two research interns (2024, 2025).

Computer Vision Researcher

Toulouse, France

THALES AVS

2021 – 2022

- Developed a Unity-based synthetic data generation pipeline using domain randomization for embedded drone detection.
- Trained and evaluated embedded drone detection models.
- Compared 3D generated datasets with GAN-based generation approaches.

Research Intern

Toulouse, France

ISAE-SUPAERO

2021

- Designed a co-evolutionary policy search strategy for video games (GECCO-W 2022).

Software Consultant

Remote, France

RTE

2020 – 2021

- Developed two open-source plugins for the FledgePOWER multi-protocol translation gateway.

Research Intern

Tsukuba, Japan

UNIVERSITY OF TSUKUBA

2019

- Proposed a procedural generation algorithm for settlements in video games.

Education

PhD in Computer Vision

Toulouse, France

ISAE-SUPAERO

2023 – June 2026

- Research topic: Synthetic-to-Real Image-Based Domain Adaptation for Robust Object Recognition.
- PhD in collaboration with Airbus.
- Advisors: Thomas Oberlin, Dennis G. Wilson, Meriem Ghrib, Manon Huguenin.

MSc in Engineering

Toulouse, France

ENSEEIH

2019 – 2022

- Major in Computer Science for Multimedia.
- Software engineering, artificial intelligence, computer vision including 3D reconstruction, camera calibration.

University Degree in Technology

Bordeaux, France

UNIVERSITY OF BORDEAUX

2017 – 2019

- Major in Computer Science.

Publications

Style Transfer with Diffusion Models for Synthetic-to-Real Domain Adaptation,

Estelle Chigot, Dennis G. Wilson, Meriem Ghrib, Thomas Oberlin. *Computer Vision and Image Understanding*, 2025.

Multi-Objective Optimization for Synthetic-to-Real Style Transfer,

Estelle Chigot, Thomas Oberlin, Manon Huguenin, Dennis G. Wilson. *International Conference on the Applications of Evolutionary Computation (Part of EvoStar)*, 2026.

Synthetic Data for Robust Runway Detection,

Estelle Chigot, Dennis G. Wilson, Meriem Ghrib, Fabrice Jimenez, Thomas Oberlin. *International Conference on Computer Analysis of Images and Patterns*, 2025.

Enhancing Runway Detection Models with Synthetic-to-Real Style Transfer,

Estelle Chigot, Solène Barrat, Manon Huguenin, Dennis G. Wilson and Thomas Oberlin. *Under review*.

Coevolution of neural networks for agents and environments,

Estelle Chigot and Dennis G. Wilson. *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, 2022.

Teaching

ISAE-SUPAERO

- **Computer Vision – Image Generation** (3h) MSc 2
- **Machine Learning Hackaton** (44h) MSc 2
- **Evolutionary Computation** (15h) MSc 1
- **Bash & Python** (15h) MSc 2

ENSEEIH

- **Object-Oriented Programming** (48h) BSc 3

Skills

Languages: French (Native), English (Proficient), German (Beginner)

Programming Languages: Python, Java, C++, C#

Deep Learning: PyTorch, TensorFlow

Tools: Linux, Docker, Slurm, Bash, Git, Unity, W&B

Interests: Dance, Video games, Baking, Reading